STM32 CubeMX

1. Description

1.1. Project

Project Name	LR111x_L552VE
Board Name	custom
Generated with:	STM32CubeMX 6.1.1
Date	02/12/2021

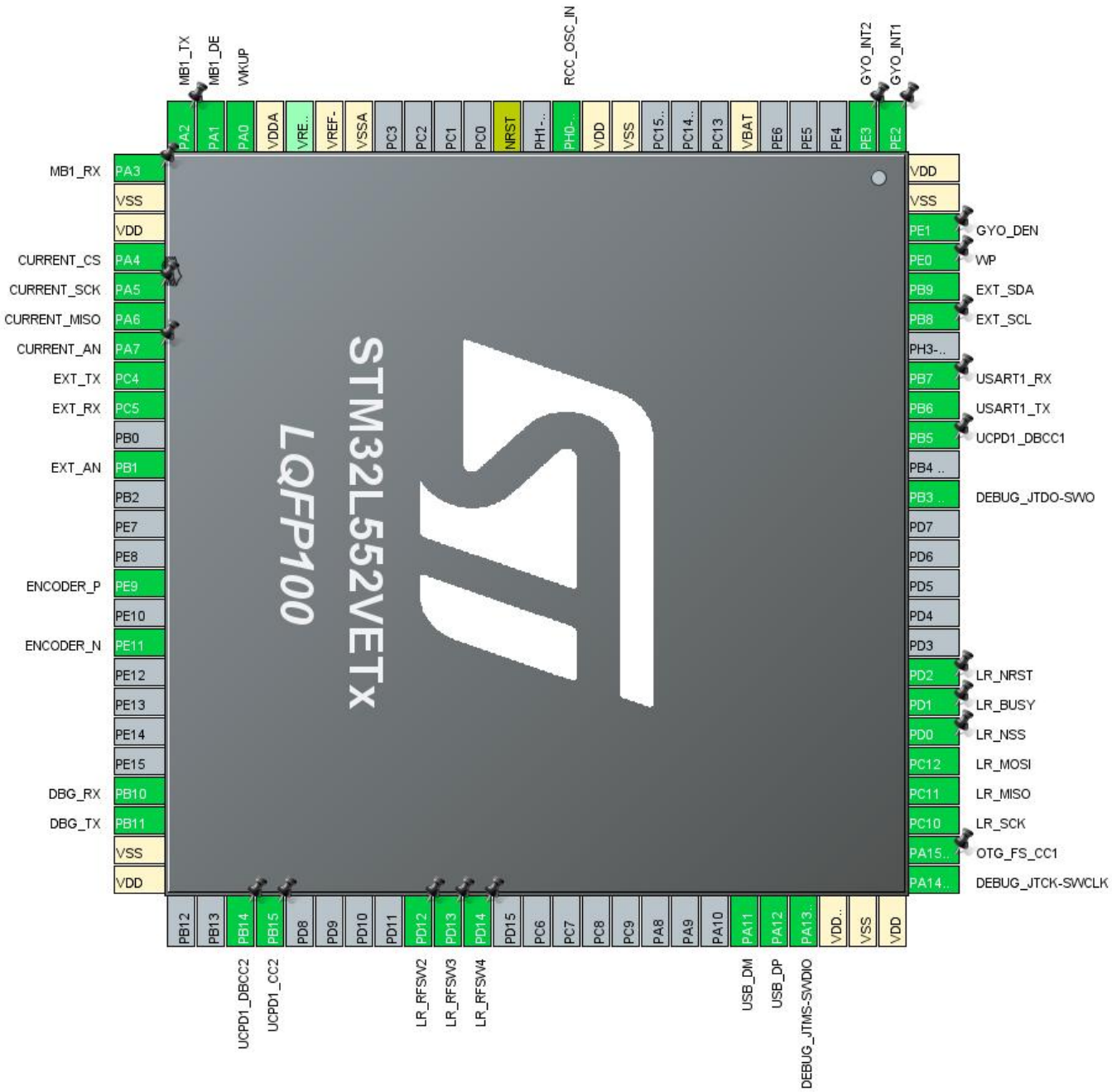
1.2. MCU

MCU Series	STM32L5
MCU Line	STM32L5x2
MCU name	STM32L552VETx
MCU Package	LQFP100
MCU Pin number	100

1.3. Core(s) information

Core(s)	ARM Cortex-M33

2. Pinout Configuration



(Rotated -270°)

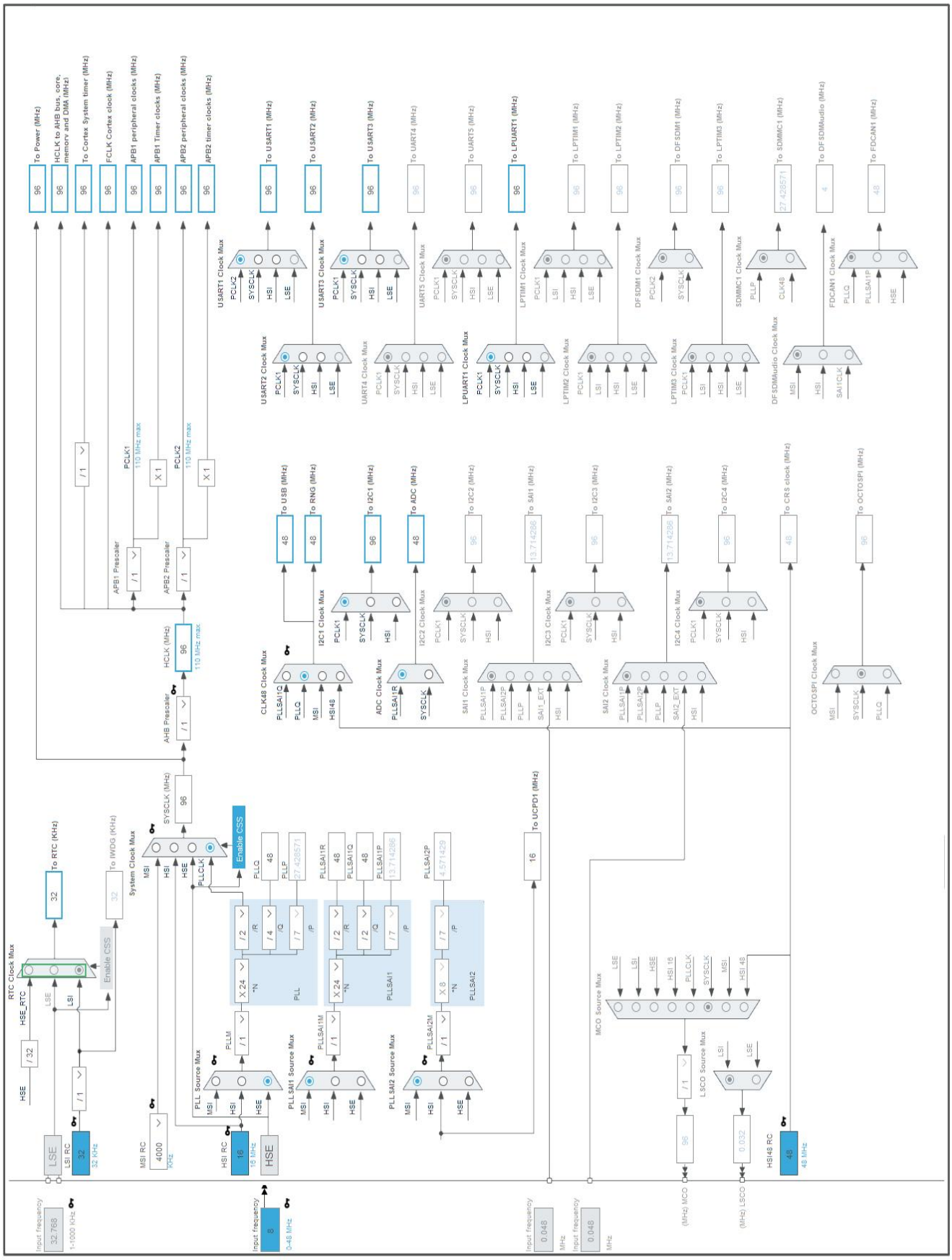
3. Pins Configuration

Pin Number LQFP100	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
1	PE2	I/O	GPIO_EXTI2	GYO_INT1
2	PE3	I/O	GPIO_EXTI3	GYO_INT2
6	VBAT	Power		
10	VSS	Power		
11	VDD	Power		
12	PH0-OSC_IN (PH0)	I/O	RCC_OSC_IN	
14	NRST	Reset		
19	VSSA	Power		
20	VREF-	Power		
22	VDDA	Power		
23	PA0	I/O	PWR_WKUP1	WKUP
24	PA1	I/O	USART2_DE	MB1_DE
25	PA2	I/O	USART2_TX	MB1_TX
26	PA3	I/O	USART2_RX	MB1_RX
27	VSS	Power		
28	VDD	Power		
29	PA4	I/O	GPIO_Output, SPI1_NSS	CURRENT_CS
30	PA5	I/O	SPI1_SCK	CURRENT_SCK
31	PA6	I/O	SPI1_MISO	CURRENT_MISO
32	PA7	I/O	ADC2_IN12	CURRENT_AN
33	PC4	I/O	USART3_TX	EXT_TX
34	PC5	I/O	USART3_RX	EXT_RX
36	PB1	I/O	ADC1_IN16	EXT_AN
40	PE9	I/O	TIM1_CH1	ENCODER_P
42	PE11	I/O	TIM1_CH2	ENCODER_N
47	PB10	I/O	LPUART1_RX	DBG_RX
48	PB11	I/O	LPUART1_TX	DBG_TX
49	VSS	Power		
50	VDD	Power		
53	PB14	I/O	UCPD1_DBCC2	
54	PB15	I/O	UCPD1_CC2	
59	PD12	I/O	GPIO_EXTI12	LR_RFSW2
60	PD13	I/O	GPIO_EXTI13	LR_RFSW3
61	PD14	I/O	GPIO_EXTI14	LR_RFSW4
70	PA11	I/O	USB_DM	
71	PA12	I/O	USB_DP	

Pin Number LQFP100	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
72	PA13 (JTMS/SWDIO)	I/O	DEBUG_JTMS-SWDIO	
73	VDDUSB	Power		
74	VSS	Power		
75	VDD	Power		
76	PA14 (JTCK/SWCLK)	I/O	DEBUG_JTCK-SWCLK	
77	PA15 (JTDI)	I/O	UCPD1_CC1	OTG_FS_CC1
78	PC10	I/O	SPI3_SCK	LR_SCK
79	PC11	I/O	SPI3_MISO	LR_MISO
80	PC12	I/O	SPI3_MOSI	LR_MOSI
81	PD0 *	I/O	GPIO_Output	LR_NSS
82	PD1	I/O	GPIO_EXTI1	LR_BUSY
83	PD2 *	I/O	GPIO_Output	LR_NRST
89	PB3 (JTDO/TRACESWO)	I/O	DEBUG_JTDO-SWO	
91	PB5	I/O	UCPD1_DBCC1	
92	PB6	I/O	USART1_TX	
93	PB7	I/O	USART1_RX	
95	PB8	I/O	I2C1_SCL	EXT_SCL
96	PB9	I/O	I2C1_SDA	EXT_SDA
97	PE0 *	I/O	GPIO_Output	WP
98	PE1 *	I/O	GPIO_Output	GYO_DEN
99	VSS	Power		
100	VDD	Power		

* The pin is affected with an I/O function

4. Clock Tree Configuration



5. Software Project

5.1. Project Settings

Name	Value
Project Name	LR111x_L552VE
Project Folder	F:\Documents\EAGLE\projects\LoRa\eMOD\LR111x\FW
Toolchain / IDE	STM32CubeIDE
Firmware Package Name and Version	STM32Cube FW_L5 V1.3.1
Application Structure	Advanced
Generate Under Root	Yes
Do not generate the main()	No
Minimum Heap Size	0x200
Minimum Stack Size	0x400

5.2. Code Generation Settings

Name	Value
STM32Cube MCU packages and embedded software	Copy only the necessary library files
Generate peripheral initialization as a pair of '.c/.h' files	No
Backup previously generated files when re-generating	Yes
Keep User Code when re-generating	Yes
Delete previously generated files when not re-generated	Yes
Set all free pins as analog (to optimize the power consumption)	No
Enable Full Assert	Yes

5.3. Advanced Settings - Generated Function Calls CortexM33S

Rank	Function Name	Peripheral Instance Name
1	MX_GPIO_Init	GPIO
2	MX_DMA_Init	DMA
3	SystemClock_Config	RCC
4	MX_GTZC_Init	GTZC
5	MX_RTC_Init	RTC

5.4. Advanced Settings - Generated Function Calls CortexM33NS

Rank	Function Name	Peripheral Instance Name
1	MX_GPIO_Init	GPIO

Rank	Function Name	Peripheral Instance Name
2	MX_DMA_Init	DMA
3	SystemClock_Config	RCC
4	MX_ADC1_Init	ADC1
5	MX_CRC_Init	CRC
6	MX_HASH_Init	HASH
7	MX_I2C1_Init	I2C1
8	MX_ICACHE_Init	ICACHE
9	MX_LPUART1_UART_Init	LPUART1
10	MX_USART1_UART_Init	USART1
11	MX_USART2_UART_Init	USART2
12	MX_USART3_UART_Init	USART3
13	MX_RNG_Init	RNG
14	MX_RTC_Init	RTC
15	MX_SPI1_Init	SPI1
16	MX_SPI3_Init	SPI3
17	MX_TIM1_Init	TIM1
18	MX_UCPD1_Init	UCPD1
19	MX_USB_Device_Init	USB_DEVICE
20	MX_ADC2_Init	ADC2
21	MX_USBPD_Init	USBPD
22	MX_TRACER_EMB_Init	TRACER_EMB

6. Power Consumption Calculator report

6.1. Microcontroller Selection

Series	STM32L5
Line	STM32L5x2
MCU	STM32L552VETx
Datasheet	DS12737_Rev5

6.2. Parameter Selection

Temperature	25
Vdd	3.0

6.3. Battery Selection

Battery	Li-SOCL2(A3400)
Capacity	3400.0 mAh
Self Discharge	0.08 %/month
Nominal Voltage	3.6 V
Max Cont Current	100.0 mA
Max Pulse Current	200.0 mA
Cells in series	1
Cells in parallel	1

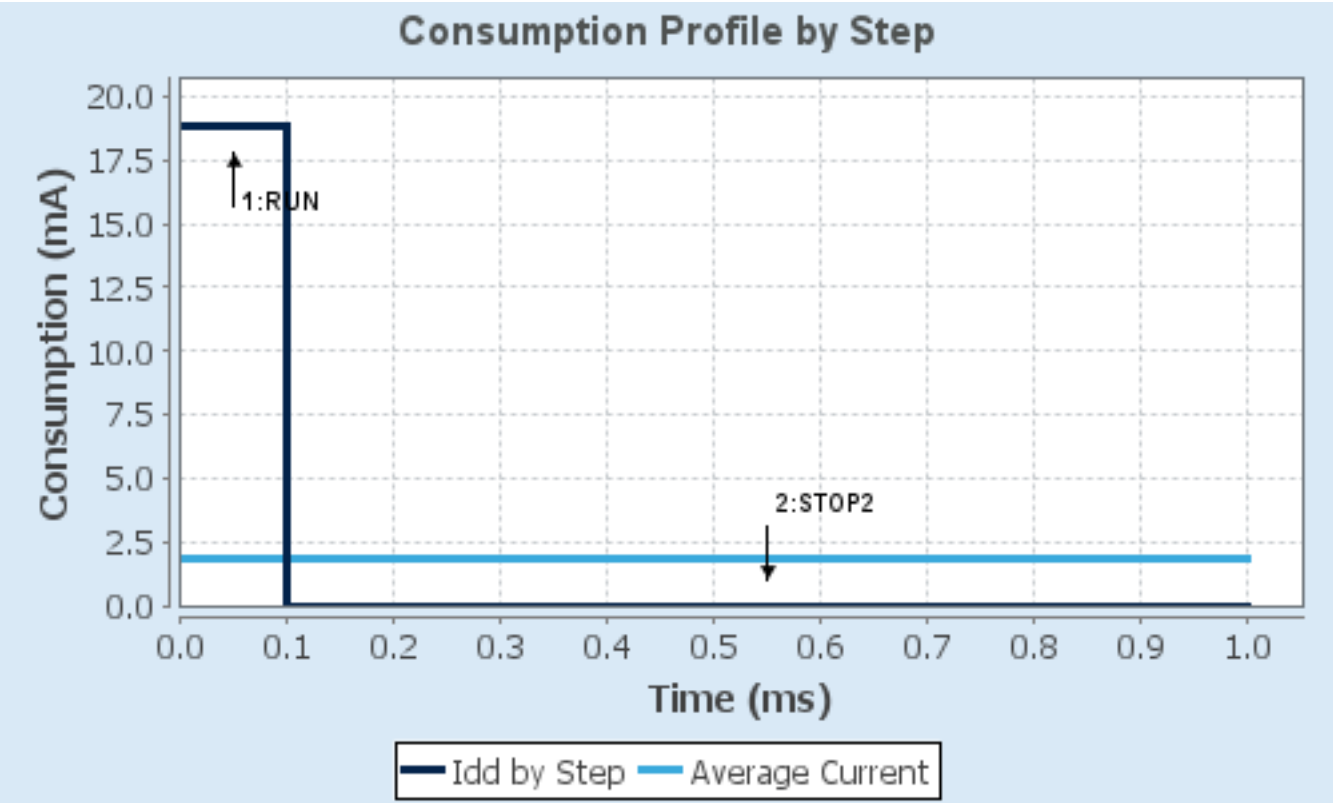
6.4. Sequence

Step	Step1	Step2
Mode	RUN	STOP2
Vdd	3.0	3.0
Voltage Source	Battery	Battery
Range	Range0-High	NoRange
Fetch Type	FLASH/SingleBank	NA
CPU Frequency	110 MHz	0 Hz
Clock Configuration	HSE BYP PLL	ALL CLOCKS OFF
Clock Source Frequency	4 MHz	0 Hz
Peripherals		
Additional Cons.	0 mA	0 mA
Average Current	18.8 mA	3.83 µA
Duration	0.1 ms	0.9 ms
DMIPS	138.0	0.0
Ta Max	102.18	105
Category	In DS Table	In DS Table

6.5. Results

Sequence Time	1 ms	Average Current	1.88 mA
Battery Life	2 months, 14 days, 5 hours	Average DMIPS	137.5 DMIPS

6.6. Chart


Consumption Profile by Step
Consumption (mA)
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
1:RUN
2:STOP2
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
Time (ms)
Idd by Step
Average Current

7. Peripherals and Middlewares Configuration

7.1. ADC1

mode: IN16 Single-ended

mode: Temperature Sensor Channel

mode: Vbat Channel

mode: Vrefint Channel

7.1.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M33 non secure

ADCs_Common_Settings:

Mode	Independent mode

ADC_Settings:

Clock Prescaler	Asynchronous clock mode divided by 1
Resolution	ADC 12-bit resolution
Data Alignment	Right alignment
Scan Conversion Mode	Disabled
Continuous Conversion Mode	Disabled
Discontinuous Conversion Mode	Disabled
DMA Continuous Requests	Disabled
End Of Conversion Selection	End of single conversion
Overrun behaviour	Overrun data preserved
Low Power Auto Wait	Disabled

ADC_Regular_ConversionMode:

Enable Regular Conversions	Enable
Enable Regular Oversampling	Disable
Number Of Conversion	1
External Trigger Conversion Source	Regular Conversion launched by software
External Trigger Conversion Edge	None
Rank	1
Channel	**Channel Vbat ***
Sampling Time	2.5 Cycles
Offset Number	No offset

ADC_Injected_ConversionMode:

Enable Injected Conversions	Disable

Analog Watchdog 1:

Enable Analog WatchDog1 Mode	false

Analog Watchdog 2:

Enable Analog WatchDog2 Mode	false

Analog Watchdog 3:

Enable Analog WatchDog3 Mode	false

7.2. ADC2

IN12: IN12 Single-ended

7.2.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M33 non secure

ADCs_Common_Settings:

Mode	Independent mode

ADC_Settings:

Clock Prescaler	Asynchronous clock mode divided by 1
Resolution	ADC 12-bit resolution
Data Alignment	Right alignment
Scan Conversion Mode	Disabled
Continuous Conversion Mode	Disabled
Discontinuous Conversion Mode	Disabled
DMA Continuous Requests	Disabled
End Of Conversion Selection	End of single conversion
Overrun behaviour	Overrun data preserved
Low Power Auto Wait	Disabled

ADC_Regular_ConversionMode:

Enable Regular Conversions	Enable
Enable Regular Oversampling	Disable
Number Of Conversion	1
External Trigger Conversion Source	Regular Conversion launched by software
External Trigger Conversion Edge	None
Rank	1
Channel	Channel 12
Sampling Time	2.5 Cycles
Offset Number	No offset

ADC_Injected_ConversionMode:

Enable Injected Conversions	Disable

Analog Watchdog 1:

Enable Analog WatchDog1 Mode	false

Analog Watchdog 2:

Enable Analog WatchDog2 Mode	false

Analog Watchdog 3:

Enable Analog WatchDog3 Mode	false

7.3. CRC

mode: Activated

7.3.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M33 non secure

Basic Parameters:

Default Polynomial State	Enable
Default Init Value State	Enable

Advanced Parameters:

Input Data Inversion Mode	None
Output Data Inversion Mode	Disable
Input Data Format	Bytes

7.4. DEBUG

Debug: Trace Asynchronous Sw

7.4.1. Core(s) Settings:

Context(s):	Cortex-M33 non secure

7.5. GTZC

mode: Enable

7.5.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M33 secure

Lock all TZSC:

Lock the configuration of TZSC (Peripherals & MPCWM) until next reset	no Peripheral & MPCWM are locked

7.5.2. TrustZone Security Controller - Securable Peripherals:

Core(s) Settings:

Context(s):	Cortex-M33 secure

Secureable Peripherals:

Configure Secure Peripheral	by Individual Securing from full Not Secured
Securable Peripheral	Secure Attribute
ADC1|2	not secured
COMP1|2	not secured
CRC	not secured
CRS	not secured
DAC1	not secured
DFSDM1	not secured
FDCAN1	not secured
FMC	not secured
HASH	not secured
I2C1	not secured
I2C2	not secured
I2C3	not secured
I2C4	not secured
ICACHE	not secured
IWDG	not secured
LPTIM1	not secured
LPTIM2	not secured
LPTIM3	not secured
LPUART1	not secured
OCTOSPI1	not secured
OPAMP1|2	not secured
RNG	not secured
SAI1	not secured
SAI2	not secured
SDMMC1	not secured
SPI1	not secured
SPI2	not secured
SPI3	not secured
TIM1	not secured
TIM2	not secured
TIM3	not secured
TIM4	not secured
TIM5	not secured
TIM6	not secured
TIM7	not secured
TIM8	not secured

TIM15	not secured
TIM16	not secured
TIM17	not secured
TSC	not secured
USART1	not secured
USART2	not secured
USART3	not secured
UART4	not secured
UART5	not secured
UCPD1	not secured
USB	not secured
VREFBUF	not secured
WWDG	not secured

7.5.3. TrustZone Security Controller - Privilegeable Peripherals:

Core(s) Settings:

Context(s):	Cortex-M33 secure

Privilegeable Peripherals:

Configure Privilege Peripheral	by Individual Privileging from full Not Privileged
Privilegeable Peripheral	Privilege Attribute
ADC1|2	not privileged
COMP1|2	not privileged
CRC	not privileged
CRS	not privileged
DAC1	not privileged
DFSDM1	not privileged
FDCAN1	not privileged
FMC	not privileged
HASH	not privileged
I2C1	not privileged
I2C2	not privileged
I2C3	not privileged
I2C4	not privileged
ICACHE_REG	not privileged
IWDG	not privileged
LPTIM1	not privileged
LPTIM2	not privileged
LPTIM3	not privileged
LPUART1	not privileged
OCTOSPI1	not privileged
OPAMP1|2	not privileged

RNG	not privileged
SAI1	not privileged
SAI2	not privileged
SDMMC1	not privileged
SPI1	not privileged
SPI2	not privileged
SPI3	not privileged
TIM1	not privileged
TIM2	not privileged
TIM3	not privileged
TIM4	not privileged
TIM5	not privileged
TIM6	not privileged
TIM7	not privileged
TIM8	not privileged
TIM15	not privileged
TIM16	not privileged
TIM17	not privileged
TSC	not privileged
USART1	not privileged
USART2	not privileged
USART3	not privileged
UART4	not privileged
UART5	not privileged
UCPD1	not privileged
USBFS	not privileged
VREFBUF	not privileged
WWDG	not privileged

7.5.4. TrustZone Security Controller - Memory Protection Controller WaterMark:

Core(s) Settings:

Context(s):	Cortex-M33 secure

MPCWM1 (OCTOSPI):

Configure Memory	from full Secured
Non Secure Area 1 Start Address	**0x90000000 ***
Non Secure Area 1 Size	**0x0 ***
Non Secure Area 2 Start Address	**0x90000000 ***
Non Secure Area 2 Size	**0x0 ***

MPCWM2 (FMC_NOR):

Configure Memory	from full Secured

Non Secure Area 1 Start Address	**0x60000000 ***
Non Secure Area 1 Size	**0x0 ***
Non Secure Area 2 Start Address	**0x60000000 ***
Non Secure Area 2 Size	**0x0 ***

MPCWM3 (FMC_NAND):

Configure Memory	from full Secured
Non Secure Area Start Address	**0x80000000 ***
Non Secure Area Size	**0x0 ***

7.5.5. Block-Based Memory Protection Controller:

Core(s) Settings:

Context(s):	Cortex-M33 secure

MPCBB1 (SRAM1):

Configuration Settings	
Authorize Secure Read/Write on Non-secure area	Secure Read/Write access allowed on non-secure SRAM block
Default security state	Default state (source clock secured if a secure area exists in the MPCBB and vice-versa)
Lock	MPCBB configuration settings are not locked
Memory Attributes Settings	
Configure Memory	Use default memory template from partition_stm32l5xx.h
Non Secure Area Start Address	**0x20018000 ***
Non Secure Area Size	**0x18000 ***
Lock	MPCBB memory attribute settings are not locked

MPCBB2 (SRAM2):

Configuration Settings	
Authorize Secure Read/Write on Non-secure area	Secure Read/Write access allowed on non-secure SRAM block
Default security state	Default state (source clock secured if a secure area exists in the MPCBB and vice-versa)
Lock	MPCBB configuration settings are not locked
Memory Attributes Settings	
Configure Memory	Use default memory template from partition_stm32l5xx.h
Non Secure Area Start Address	**0x20030000 ***
Non Secure Area Size	**0x10000 ***
Lock	MPCBB memory attribute settings are not locked

7.5.6. TrustZone Illegal access Controller:

Core(s) Settings:

Context(s): Cortex-M33 secure

Illegal Access ITs for securable & TZ-Aware Peripherals:

Configure Illegal access IT	by Individual Enabling from full Disabled
Illegal Access Interrupt Peripheral	Illegal Access Interrupt State
ADC1|2	disabled
COMP1|2	disabled
CRS	disabled
CRS	disabled
DAC1	disabled
DFSDM1	disabled
DMA1	disabled
DMA2	disabled
DMAMUX1	disabled
EXTI	disabled
FDCAN1	disabled
FLASH	disabled
FLASH_REG	disabled
FMC_MEM	disabled
FMC_REG	disabled
HASH	disabled
I2C1	disabled
I2C2	disabled
I2C3	disabled
I2C4	disabled
ICACHE_REG	disabled
IWDG	disabled
LPTIM1	disabled
LPTIM2	disabled
LPTIM3	disabled
LPUART1	disabled
MPCBB1_REG	disabled
MPCBB2_REG	disabled
OCTOSPI1_MEM	disabled
OCTOSPI1_REG	disabled
OPAMP	disabled
PWR	disabled
RCC	disabled
RNG	disabled
RTC	disabled
SAI1	disabled
SAI2	disabled
SDMMC1	disabled

SPI1	disabled
SPI2	disabled
SPI3	disabled
SRAM1	disabled
SRAM2	disabled
SYS	disabled
TIM1	disabled
TIM2	disabled
TIM3	disabled
TIM4	disabled
TIM5	disabled
TIM6	disabled
TIM7	disabled
TIM8	disabled
TIM15	disabled
TIM16	disabled
TIM17	disabled
TSC	disabled
TZIC	disabled
TZSC	disabled
USART1	disabled
USART2	disabled
USART3	disabled
UART4	disabled
UART5	disabled
UCPD1	disabled
USBFS	disabled
VREFBUF	disabled
WWDG	disabled

7.6. HASH

mode: Activated

7.6.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M33 non secure

Algorithm:

Secure hash algorithm type	SHA1

Parameters:

Hash data type in bit	32

7.7. I2C1

I2C: I2C

7.7.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M33 non secure

Timing configuration:

Custom Timing	Disabled
I2C Speed Mode	Standard Mode
I2C Speed Frequency (KHz)	100
Rise Time (ns)	0
Fall Time (ns)	0
Coefficient of Digital Filter	0
Analog Filter	Enabled
Timing	**0x10B0DCFB ***

Slave Features:

Clock No Stretch Mode	Disabled
General Call Address Detection	Disabled
Primary Address Length selection	7-bit
Dual Address Acknowledged	Disabled
Primary slave address	0

7.8. ICACHE

Mode: 2-ways set associative cache

Secure Attribute: not secured

7.8.1. Core(s) Settings:

Context(s):	Cortex-M33 non secure

7.9. LPUART1

Mode: Multiprocessor Communication

7.9.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M33 non secure

Basic Parameters:

Baud Rate	209700
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Single Sample	Disable
Wake-Up Method	Idle Line
ClockPrescaler	1
Fifo Mode	FIFO mode disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

7.10. PWR

mode: Wake-Up 0

Power Voltage Detector In: Power Voltage Detector In (Internal analog voltage)

mode: save power of non-active UCPD - deactive Dead Battery pull-up

7.10.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M33 secure Cortex-M33 non secure

Programmable_Voltage_Detector_Settings:

PVD detection Level	PWR PVD LEVEL 0 (2.0 V)
PWR PVD Mode	basic mode is used

7.10.2. TZSC Privilegeable PWR:

Core(s) Settings:

Context(s):	Cortex-M33 secure Cortex-M33 non secure

Programmable_Voltage_Detector_Settings:

PWR Full Privilege	Disable

7.10.3. TZSC Securable PWR:

Core(s) Settings:

Context(s):	Cortex-M33 secure Cortex-M33 non secure

Programmable_Voltage_Detector_Settings:

PWR full secure	Enable

7.11. RCC

High Speed Clock (HSE): BYPASS Clock Source

7.11.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M33 secure Cortex-M33 non secure

System Parameters:

VDD voltage (V)	3.3
Flash Latency(WS)	4 WS (5 CPU cycle)

RCC Parameters:

HSI Calibration Value	64
MSI Calibration Value	0
MSI Auto Calibration	Disabled
HSE Startup Timout Value (ms)	100
LSE Startup Timout Value (ms)	5000

Power Parameters:

Power Regulator Voltage Scale	Power Regulator Voltage Scale 0

7.12. RNG

mode: Activated

7.12.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M33 non secure
Clock Error Detection	Enable

7.13. RTC

mode: Activate Clock Source

mode: Activate Calendar

7.13.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M33 secure Cortex-M33 non secure

General:

Hour Format	Hourformat 24
Asynchronous Predivider value	127
Synchronous Predivider value	255

Calendar Time:

Data Format	BCD data format
Hours	0
Minutes	0
Seconds	0
Day Light Saving: value of hour adjustment	Daylightsaving None
Store Operation	Storeoperation Reset

Calendar Date:

Week Day	Monday
Month	January
Date	1
Year	0

7.13.2. TZSC Securable RTC:

Core(s) Settings:

Context(s):	Cortex-M33 secure Cortex-M33 non secure

Secure RTC:

RTC full secure	Enable

Backup register:

Start zone 1	RTC_BKP_DR0
Start zone 2	RTC_BKP_DR0
Start zone 3	RTC_BKP_DR0

7.13.3. TZSC Privilegeable RTC:

Core(s) Settings:

Context(s):	Cortex-M33 secure Cortex-M33 non secure

Privilege RTC:

RTC full privilege	Disable

Backup register:

Start zone 1	RTC_BKP_DR0
Start Zone 2	RTC_BKP_DR0
start zone 3	RTC_BKP_DR0

Privilege Backup register :

Backup Register PrivZone	Non-privilege

Privilege RTC Feature:

RTC_FEATURE_INIT	Non-Privilege
RTC_FEATURE_ALRA	Non-Privilege
RTC_FEATURE_ALRB	Non-Privilege
RTC_FEATURE_CAL	Non-Privilege
RTC_FEATURE_TS	Non-Privilege
RTC_FEATURE_WUT	Non-Privilege

7.14. SPI1

Mode: Receive Only Master

7.14.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M33 non secure
Basic Parameters:	
Frame Format	Motorola
Data Size	4 Bits
First Bit	MSB First
Clock Parameters:	
Prescaler (for Baud Rate)	**4 ***
Baud Rate	**24.0 MBits/s ***
Clock Polarity (CPOL)	Low
Clock Phase (CPHA)	1 Edge
Advanced Parameters:	
CRC Calculation	Disabled
NSSP Mode	Enabled
NSS Signal Type	Software

7.15. SPI3

Mode: Full-Duplex Master

7.15.1. Parameter Settings:

Core(s) Settings:	
Context(s):	Cortex-M33 non secure
Basic Parameters:	
Frame Format	Motorola
Data Size	4 Bits
First Bit	MSB First
Clock Parameters:	
Prescaler (for Baud Rate)	**4 ***
Baud Rate	**24.0 MBits/s ***
Clock Polarity (CPOL)	Low
Clock Phase (CPHA)	1 Edge
Advanced Parameters:	
CRC Calculation	Disabled
NSSP Mode	Enabled
NSS Signal Type	Software

7.16. SYS

Timebase Source: TIM6

7.16.1. Core(s) Settings:

Context(s):	Cortex-M33 non secure

7.17. SYS_S

Timebase Source: SysTick

7.17.1. Core(s) Settings:

Context(s):	Cortex-M33 secure

7.18. TIM1

Combined Channels: Encoder Mode

7.18.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M33 non secure

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Counter Period (AutoReload Register - 16 bits value)	65535
Internal Clock Division (CKD)	No Division
Repetition Counter (RCR - 16 bits value)	0
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)
Trigger Event Selection TRGO2	Reset (UG bit from TIMx_EGR)

Encoder:

Encoder Mode	Encoder Mode TI1
____ Parameters for Channel 1 ____	
Polarity	Rising Edge
IC Selection	Direct
Prescaler Division Ratio	No division
Input Filter	0

____ Parameters for Channel 2 ____

Polarity	Rising Edge
IC Selection	Direct
Prescaler Division Ratio	No division
Input Filter	0

7.19. TRACER_EMB
Uart Trace Source: USART1
7.19.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M33 non secure
Version	TRACER_EMB
TRACER_EMB request USART1 TX DMA	enabled

7.20. UCPD1
UCPD Mode: Dual Role
mode: Dead Battery Signals
7.20.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M33 non secure
Version	1.0

7.21. USART1
Mode: Asynchronous
7.21.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M33 non secure

Basic Parameters:

Baud Rate	921600
Word Length	8 Bits (including Parity)
Parity	None

Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	Disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

7.22. USART2

Mode: Asynchronous

mode: Hardware Flow Control (RS485)

7.22.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M33 non secure

Basic Parameters:

Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	Disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Polarity	High
Assertion Time	0
Deassertion Time	0

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

7.23. USART3

Mode: Asynchronous

7.23.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M33 non secure

Basic Parameters:

Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	Disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable

DMA on RX Error	Enable
MSB First	Disable

7.24. USB
mode: Device (FS)
7.24.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M33 non secure

Basic Parameters:

Speed	Full Speed 12MBit/s
Physical interface	Internal Phy
Sof Enable	Disabled

Power Parameters:

Low Power	Disabled
Link Power Management	Disabled
Battery Charging	**Enabled ***

7.25. FREERTOS
Interface: CMSIS_V2
7.25.1. Config parameters:

Core(s) Settings:

Context(s):	Cortex-M33 non secure

API:

FreeRTOS API	CMSIS v2

Versions:

FreeRTOS version	10.2.1
CMSIS-RTOS version	2.00

Security settings:

ENABLE_TRUSTZONE	Enabled
RUN_FREERTOS_SECURE_ONLY	Disabled

MPU/FPU:

ENABLE_MPU	Disabled
ENABLE_FPU	Disabled

Kernel settings:

MINIMAL_SECURE_STACK_SIZE	512
USE_PREEMPTION	Enabled
CPU_CLOCK_HZ	SystemCoreClock
TICK_RATE_HZ	1000
MAX_PRIORITIES	56
MINIMAL_STACK_SIZE	128
MAX_TASK_NAME_LEN	16
USE_16_BIT_TICKS	Disabled
IDLE_SHOULD_YIELD	Enabled
USE_MUTEXES	Enabled
USE_RECURSIVE_MUTEXES	Enabled
USE_COUNTING_SEMAPHORES	Enabled
QUEUE_REGISTRY_SIZE	8
USE_APPLICATION_TASK_TAG	Disabled
ENABLE_BACKWARD_COMPATIBILITY	Enabled
USE_PORT_OPTIMISED_TASK_SELECTION	Disabled
USE_TICKLESS_IDLE	Disabled
USE_TASK_NOTIFICATIONS	Enabled
RECORD_STACK_HIGH_ADDRESS	Disabled

Memory management settings:

Memory Allocation	Dynamic / Static
TOTAL_HEAP_SIZE	8192
Memory Management scheme	heap_4

Hook function related definitions:

USE_IDLE_HOOK	Disabled
USE_TICK_HOOK	Disabled
USE_MALLOC_FAILED_HOOK	Disabled
USE_DAEMON_TASK_STARTUP_HOOK	Disabled
CHECK_FOR_STACK_OVERFLOW	Disabled

Run time and task stats gathering related definitions:

GENERATE_RUN_TIME_STATS	Disabled
USE_TRACE_FACILITY	Enabled
USE_STATS_FORMATTING_FUNCTIONS	Disabled

Co-routine related definitions:

USE_CO_ROUTINES	Disabled
MAX_CO_ROUTINE_PRIORITIES	2

Software timer definitions:

USE_TIMERS	Enabled
TIMER_TASK_PRIORITY	2
TIMER_QUEUE_LENGTH	10
TIMER_TASK_STACK_DEPTH	256

Interrupt nesting behaviour configuration:

LIBRARY_LOWEST_INTERRUPT_PRIORITY	7
LIBRARY_MAX_SYSCALL_INTERRUPT_PRIORITY	5

Added with 10.2.1 support:

MESSAGE_BUFFER_LENGTH_TYPE	size_t
USE_POSIX_ERRNO	Disabled

7.25.2. Include parameters:

Core(s) Settings:

Context(s): Cortex-M33 non secure

Include definitions:

vTaskPrioritySet	Enabled
uxTaskPriorityGet	Enabled
vTaskDelete	Enabled
vTaskCleanUpResources	Disabled
vTaskSuspend	Enabled
vTaskDelayUntil	Enabled
vTaskDelay	Enabled
xTaskGetSchedulerState	Enabled
xTaskResumeFromISR	Enabled
xQueueGetMutexHolder	Enabled
xSemaphoreGetMutexHolder	Disabled
pcTaskGetTaskName	Disabled
uxTaskGetStackHighWaterMark	Enabled
xTaskGetCurrentTaskHandle	Disabled
eTaskGetState	Enabled
xEventGroupSetBitFromISR	Disabled
xTimerPendFunctionCall	Enabled
xTaskAbortDelay	Disabled
xTaskGetHandle	Disabled
uxTaskGetStackHighWaterMark2	Disabled

7.25.3. Advanced settings:

Core(s) Settings:

Context(s): Cortex-M33 non secure

Newlib settings (see parameter description first):

USE_NEWLIB_REENTRANT	Disabled

Project settings:

Use FW pack heap file	Enabled

7.26. USBPD

mode: Port Configuration

Stack Configuration: PD3 Full Stack

Timer service Source: TIM2

7.26.1. Parameter Settings:

Core(s) Settings:	
Context(s):	Cortex-M33 non secure
USBPD Versions:	
Core	3.0.2
Device	2.0.1
USBPD needs:	
USBPD request UCPD1 NVIC	enabled
USBPD request UCPD1 DMA	enabled

7.26.2. DPM Core Parameters:

Core(s) Settings:	
Context(s):	Cortex-M33 non secure
USB IF and Manufacturer ID:	
Vendor ID	0x0483
Product ID	0x0002
XID	0xF0000003

7.26.3. Stack Port 0 Parameters:

Core(s) Settings:	
Context(s):	Cortex-M33 non secure
Port Configuration:	
UCPD Instance	UCPD1
DMA Request RX for UCPD Port 0	UCPD1_RX_DMA1_Channel_1
DMA Request TX for UCPD Port 0	UCPD1_TX_DMA1_Channel_2
Start of Packet Parameters:	
SOP	Supported
SOP'	Not supported

SOP"	Not supported
SOP' debug	Not supported
SOP" debug	Not supported

Port 0 Parameters:

Specification revision value	Revision 2.0
Default port role	Sink
Port role swap	Supported
Data role swap to DFP	Supported
Data role swap to UFP	Supported
Vendor defined messages	Not supported
Discover Identity response	Not supported
Discover Identity sent	Not supported
Ping	Not supported
Caps counter	Not supported

Cable Detection Parameters:

CAD role toggle	Not supported
CAD default resistor	Default USB Power
CAD accessory	Not supported

7.26.4. PDO General Definitions:

Core(s) Settings:

Context(s):	Cortex-M33 non secure
Number of Source PDOs for port 0	1
port 0 Source PDO 1	**0x2201912C ***
Number of Sink PDOs for port 0	1
port 0 Sink PDO 1	**0x26019096 ***

7.26.5. User Port 0 Parameters:

Core(s) Settings:

Context(s):	Cortex-M33 non secure

Port 0 Parameters:

Data role swap	Not supported
VCONN swap	Not supported

7.27. USB_DEVICE

Class For FS IP: Communication Device Class (Virtual Port Com)

7.27.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M33 non secure

Basic Parameters:

USBD_MAX_NUM_INTERFACES (Maximum number of supported interfaces)	1
USBD_MAX_NUM_CONFIGURATION (Maximum number of supported configuration)	1
USBD_MAX_STR_DESC_SIZ (Maximum size for the string descriptors)	512
USBD_SELF_POWERED (Enabled self power)	Enabled
USBD_DEBUG_LEVEL (USBD Debug Level)	**2: User + Error messages ***
USBD_LPM_ENABLED (Link Power Management)	1: Link Power Management supported

Class Parameters:

USB CDC Rx Buffer Size	2048
USB CDC Tx Buffer Size	2048

7.27.2. Device Descriptor:

Core(s) Settings:

Context(s):	Cortex-M33 non secure

Device Descriptor:

VID (Vendor IDentifier)	1155
LANGID_STRING (Language Identifier)	English(United States)
MANUFACTURER_STRING (Manufacturer Identifier)	STMicroelectronics

Device Descriptor FS:

PID (Product IDentifier)	22336
PRODUCT_STRING (Product Identifier)	STM32 Virtual ComPort
CONFIGURATION_STRING (Configuration Identifier)	CDC Config
INTERFACE_STRING (Interface Identifier)	CDC Interface

*** User modified value**

8. System Configuration

8.1. GPIO configuration

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label	Context
ADC1	PB1	ADC1_IN16	Analog mode	No pull-up and no pull-down	n/a	EXT_AN	Cortex-M33 non secure
ADC2	PA7	ADC2_IN12	Analog mode	No pull-up and no pull-down	n/a	CURRENT_AN	Cortex-M33 non secure
DEBUG	PA13 (JTMS/SWDIO)	DEBUG_JTMS-SWDIO	n/a	n/a	n/a		Cortex-M33 non secure
	PA14 (JTCK/SWCLK)	DEBUG_JTCK-SWCLK	n/a	n/a	n/a		Cortex-M33 non secure
	PB3 (JTDO/TRACESWO)	DEBUG_JTDO-SWO	n/a	n/a	n/a		Cortex-M33 non secure
I2C1	PB8	I2C1_SCL	Alternate Function Open Drain	Pull-up	Low	EXT_SCL	Cortex-M33 non secure
	PB9	I2C1_SDA	Alternate Function Open Drain	Pull-up	Low	EXT_SDA	Cortex-M33 non secure
LPUART1	PB10	LPUART1_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	DBG_RX	Cortex-M33 non secure
	PB11	LPUART1_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	DBG_TX	Cortex-M33 non secure
PWR	PA0	PWR_WKUP1	n/a	n/a	n/a	WKUP	Cortex-M33 secure* Cortex-M33 non
RCC	PH0-OSC_IN (PH0)	RCC_OSC_IN	n/a	n/a	n/a		Cortex-M33 secure* Cortex-M33 non
SPI1	PA4	SPI1_NSS	Output Push Pull	No pull-up and no pull-down	Low	CURRENT_CS	Cortex-M33 non secure
	PA5	SPI1_SCK	Alternate Function Push Pull	No pull-up and no pull-down	Low	CURRENT_SCK	Cortex-M33 non secure
	PA6	SPI1_MISO	Alternate Function Push Pull	No pull-up and no pull-down	Low	CURRENT_MISO	Cortex-M33 non secure
SPI3	PC10	SPI3_SCK	Alternate Function Push Pull	No pull-up and no pull-down	Low	LR_SCK	Cortex-M33 non secure
	PC11	SPI3_MISO	Alternate Function Push Pull	No pull-up and no pull-down	Low	LR_MISO	Cortex-M33 non secure
	PC12	SPI3_MOSI	Alternate Function Push Pull	No pull-up and no pull-down	Low	LR_MOSI	Cortex-M33 non secure
TIM1	PE9	TIM1_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Low	ENCODER_P	Cortex-M33 non secure

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label	Context
	PE11	TIM1_CH2	Alternate Function Push Pull	No pull-up and no pull-down	Low	ENCODER_N	Cortex-M33 non secure
UCPD1	PB14	UCPD1_DBCC2	n/a	n/a	n/a		Cortex-M33 non secure
	PB15	UCPD1_CC2	Analog mode	No pull-up and no pull-down	n/a		Cortex-M33 non secure
	PA15 (JTDI)	UCPD1_CC1	Analog mode	No pull-up and no pull-down	n/a	OTG_FS_CC1	Cortex-M33 non secure
	PB5	UCPD1_DBCC1	n/a	n/a	n/a		Cortex-M33 non secure
USART1	PB6	USART1_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M33 non secure
	PB7	USART1_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M33 non secure
USART2	PA1	USART2_DE	Alternate Function Push Pull	No pull-up and no pull-down	Low	MB1_DE	Cortex-M33 non secure
	PA2	USART2_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	MB1_TX	Cortex-M33 non secure
	PA3	USART2_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	MB1_RX	Cortex-M33 non secure
USART3	PC4	USART3_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	EXT_TX	Cortex-M33 non secure
	PC5	USART3_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	EXT_RX	Cortex-M33 non secure
USB	PA11	USB_DM	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M33 non secure
	PA12	USB_DP	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M33 non secure
GPIO	PE2	GPIO_EXTI2	External Interrupt Mode with Rising edge trigger detection	No pull-up and no pull-down	n/a	GYO_INT1	
	PE3	GPIO_EXTI3	External Interrupt Mode with Rising edge trigger detection	No pull-up and no pull-down	n/a	GYO_INT2	
	PA4	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	CURRENT_CS	
	PD12	GPIO_EXTI12	External Interrupt Mode with Rising edge trigger detection	No pull-up and no pull-down	n/a	LR_RFSW2	
	PD13	GPIO_EXTI13	External Interrupt Mode with Rising edge trigger detection	No pull-up and no pull-down	n/a	LR_RFSW3	
	PD14	GPIO_EXTI14	External Interrupt Mode with Rising edge trigger detection	No pull-up and no pull-down	n/a	LR_RFSW4	
	PD0	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	LR_NSS	

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label	Context
	PD1	GPIO_EXTI1	External Interrupt Mode with Rising edge trigger detection	No pull-up and no pull-down	n/a	LR_BUSY	
	PD2	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	LR_NRST	
	PE0	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	WP	
	PE1	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	GYO_DEN	

* Initialized context

8.2. Managing security in TrustZone®-aware peripherals

- It is recommended to secure some peripheral(s).

Please take attention to "Summary of the I/Os that can be secured and connected to a non-secure peripheral" table in "General-purpose I/Os (GPIO)" paragraph of "Managing security in TrustZone®-aware peripherals" section in Reference Manual

8.3. DMA configuration

DMA request	Stream	Direction	Priority
UCPD1_RX	DMA1_Channel1	Peripheral To Memory	Low
UCPD1_TX	DMA1_Channel2	Memory To Peripheral	Low
USART1_TX	DMA1_Channel3	Memory To Peripheral	Low
USART1_RX	DMA1_Channel4	Peripheral To Memory	Low
LPUART1_RX	DMA1_Channel5	Peripheral To Memory	Low
LPUART1_TX	DMA1_Channel6	Memory To Peripheral	Low
SPI1_RX	DMA1_Channel7	Peripheral To Memory	Low
SPI3_RX	DMA1_Channel8	Peripheral To Memory	Low
SPI3_TX	DMA2_Channel1	Memory To Peripheral	Low
USART3_RX	DMA2_Channel2	Peripheral To Memory	Low
USART3_TX	DMA2_Channel3	Memory To Peripheral	Low
USART2_RX	DMA2_Channel4	Peripheral To Memory	Low
USART2_TX	DMA2_Channel5	Memory To Peripheral	Low
ADC1	DMA2_Channel6	Peripheral To Memory	Low
HASH_IN	DMA2_Channel7	Memory To Peripheral	Low

UCPD1_RX: DMA1_Channel1 DMA request Settings:

Mode: Normal
Peripheral Increment: Disable
Memory Increment: **Enable ***
Peripheral Data Width: Byte
Memory Data Width: Byte

UCPD1_TX: DMA1_Channel2 DMA request Settings:

Mode: Normal
Peripheral Increment: Disable
Memory Increment: **Enable ***
Peripheral Data Width: Byte
Memory Data Width: Byte

USART1_TX: DMA1_Channel3 DMA request Settings:

Mode: Normal
Peripheral Increment: Disable
Memory Increment: **Enable ***

Peripheral Data Width:	Byte
Memory Data Width:	Byte

USART1_RX: DMA1_Channel4 DMA request Settings:

Mode:	Normal
Peripheral Increment:	Disable
Memory Increment:	**Enable ***
Peripheral Data Width:	Byte
Memory Data Width:	Byte

LPUART1_RX: DMA1_Channel5 DMA request Settings:

Mode:	Normal
Peripheral Increment:	Disable
Memory Increment:	**Enable ***
Peripheral Data Width:	Byte
Memory Data Width:	Byte

LPUART1_TX: DMA1_Channel6 DMA request Settings:

Mode:	Normal
Peripheral Increment:	Disable
Memory Increment:	**Enable ***
Peripheral Data Width:	Byte
Memory Data Width:	Byte

SPI1_RX: DMA1_Channel7 DMA request Settings:

Mode:	Normal
Peripheral Increment:	Disable
Memory Increment:	**Enable ***
Peripheral Data Width:	Byte
Memory Data Width:	Byte

SPI3_RX: DMA1_Channel8 DMA request Settings:

Mode:	Normal
Peripheral Increment:	Disable

Memory Increment:	**Enable ***
Peripheral Data Width:	Byte
Memory Data Width:	Byte

SPI3_TX: DMA2_Channel1 DMA request Settings:

Mode:	Normal
Peripheral Increment:	Disable
Memory Increment:	**Enable ***
Peripheral Data Width:	Byte
Memory Data Width:	Byte

USART3_RX: DMA2_Channel2 DMA request Settings:

Mode:	Normal
Peripheral Increment:	Disable
Memory Increment:	**Enable ***
Peripheral Data Width:	Byte
Memory Data Width:	Byte

USART3_TX: DMA2_Channel3 DMA request Settings:

Mode:	Normal
Peripheral Increment:	Disable
Memory Increment:	**Enable ***
Peripheral Data Width:	Byte
Memory Data Width:	Byte

USART2_RX: DMA2_Channel4 DMA request Settings:

Mode:	Normal
Peripheral Increment:	Disable
Memory Increment:	**Enable ***
Peripheral Data Width:	Byte
Memory Data Width:	Byte

USART2_TX: DMA2_Channel5 DMA request Settings:

Mode:	Normal

Peripheral Increment:	Disable
Memory Increment:	**Enable ***
Peripheral Data Width:	Byte
Memory Data Width:	Byte

ADC1: DMA2_Channel6 DMA request Settings:

Mode:	Normal
Peripheral Increment:	Disable
Memory Increment:	**Enable ***
Peripheral Data Width:	Half Word
Memory Data Width:	Half Word

HASH_IN: DMA2_Channel7 DMA request Settings:

Mode:	Normal
Peripheral Increment:	Disable
Memory Increment:	**Enable ***
Peripheral Data Width:	Byte
Memory Data Width:	Byte

8.4. NVIC configuration

8.4.1. NVIC1

Interrupt Table	Enable	Preenmption Priority	SubPriority
Memory management fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Pendable request for system service	true	7	0
System tick timer	true	7	0
DMA1 channel1 global interrupt	true	5	0
DMA1 channel2 global interrupt	true	5	0
DMA1 channel3 global interrupt	true	5	0
DMA1 channel4 global interrupt	true	5	0
DMA1 channel5 global interrupt	true	5	0
DMA1 channel6 global interrupt	true	5	0
DMA1 channel7 global interrupt	true	5	0
DMA1 channel8 global interrupt	true	5	0
ADC1 and ADC2 interrupts	true	5	0
TIM6 global interrupt	true	0	0
SPI1 global interrupt	true	5	0
USART1 global interrupt / USART1 wake-up interrupt through EXTI line 26	true	5	0
USART2 global interrupt / USART2 wake-up interrupt through EXTI line 27	true	5	0
USART3 global interrupt / USART3 wake-up interrupt through EXTI line 28	true	5	0
LPUART1 global interrupt / LPUART1 wake-up interrupt through EXTI line 31	true	5	0
USB FS global interrupt / USB FS wake-up interrupt through EXTI line 34	true	5	0
DMA2 channel1 global interrupt	true	5	0
DMA2 channel2 global interrupt	true	5	0
DMA2 channel3 global interrupt	true	5	0
DMA2 channel4 global interrupt	true	5	0
DMA2 channel5 global interrupt	true	5	0
DMA2 channel6 global interrupt	true	5	0
DMA2 channel7 global interrupt	true	5	0
RNG global interrupt	true	5	0
HASH global interrupt	true	5	0
SPI3 global interrupt	true	5	0
UCPD1 interrupt / UCPD1 wake-up interrupt through EXTI line 41	true	5	0
PVD/PVM1/PVM2/PVM3/PVM4 interrupts	unused		

Interrupt Table	Enable	Preenmption Priority	SubPriority
through EXTI lines 16/35/36/37/38			
Flash non-secure global interrupt	unused		
RCC non-secure global interrupt	unused		
EXTI line1 interrupt	unused		
EXTI line2 interrupt	unused		
EXTI line3 interrupt	unused		
EXTI line12 interrupt	unused		
EXTI line13 interrupt	unused		
EXTI line14 interrupt	unused		
TIM1 break interrupt	unused		
TIM1 update interrupt	unused		
TIM1 trigger and commutation interrupts	unused		
TIM1 capture compare interrupt	unused		
I2C1 event interrupt / I2C1 wake-up interrupt through EXTI line 23	unused		
I2C1 error interrupt	unused		
FPU global interrupt	unused		
Instruction cache global interrupt	unused		

8.4.2. NVIC1 Code generation

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Memory management fault	false	true	false
Undefined instruction or illegal state	false	true	false
System service call via SWI instruction	false	false	false
Pendable request for system service	false	false	false
System tick timer	false	false	true
DMA1 channel1 global interrupt	false	true	true
DMA1 channel2 global interrupt	false	true	true
DMA1 channel3 global interrupt	false	true	true
DMA1 channel4 global interrupt	false	true	true
DMA1 channel5 global interrupt	false	true	true
DMA1 channel6 global interrupt	false	true	true
DMA1 channel7 global interrupt	false	true	true
DMA1 channel8 global interrupt	false	true	true
ADC1 and ADC2 interrupts	false	true	true
TIM6 global interrupt	false	true	true
SPI1 global interrupt	false	true	true
USART1 global interrupt / USART1 wake-up interrupt through EXTI line 26	false	true	true
USART2 global interrupt / USART2 wake-	false	true	true

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
up interrupt through EXTI line 27			
USART3 global interrupt / USART3 wake-up interrupt through EXTI line 28	false	true	true
LPUART1 global interrupt / LPUART1 wake-up interrupt through EXTI line 31	false	true	true
USB FS global interrupt / USB FS wake-up interrupt through EXTI line 34	false	true	true
DMA2 channel1 global interrupt	false	true	true
DMA2 channel2 global interrupt	false	true	true
DMA2 channel3 global interrupt	false	true	true
DMA2 channel4 global interrupt	false	true	true
DMA2 channel5 global interrupt	false	true	true
DMA2 channel6 global interrupt	false	true	true
DMA2 channel7 global interrupt	false	true	true
RNG global interrupt	false	true	true
HASH global interrupt	false	true	true
SPI3 global interrupt	false	true	true
UCPD1 interrupt / UCPD1 wake-up interrupt through EXTI line 41	false	true	false

8.4.3. NVIC2

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Prefetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
Secure fault	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	0	0
System tick timer	true	0	0
Global TrustZone controller global interrupt	true	0	0
PVD/PVM1/PVM2/PVM3/PVM4 interrupts through EXTI lines 16/35/36/37/38	unused		
Flash secure global interrupt	unused		
RCC secure global interrupt	unused		
EXTI line1 interrupt	unused		
EXTI line2 interrupt	unused		
EXTI line3 interrupt	unused		
EXTI line12 interrupt	unused		

Interrupt Table	Enable	Preenmption Priority	SubPriority
EXTI line13 interrupt	unused		
EXTI line14 interrupt	unused		
FPU global interrupt	unused		

8.4.4. NVIC2 Code generation

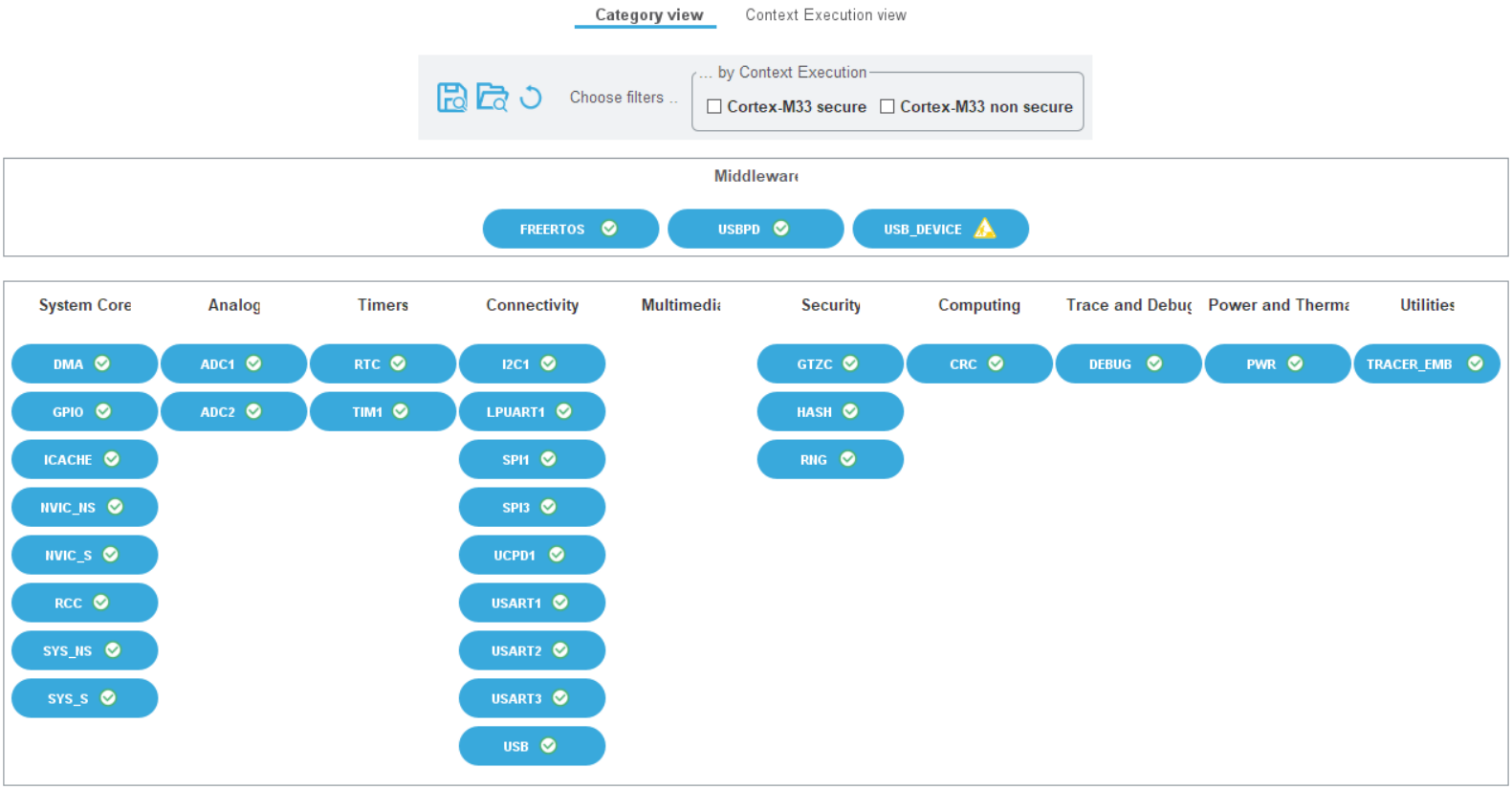
Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	false
Hard fault interrupt	false	true	false
Memory management fault	false	true	false
Prefetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false
Secure fault	false	true	false
System service call via SWI instruction	false	true	false
Debug monitor	false	true	false
Pendable request for system service	false	true	false
System tick timer	false	true	true
Global TrustZone controller global interrupt	false	true	true

*** User modified value**

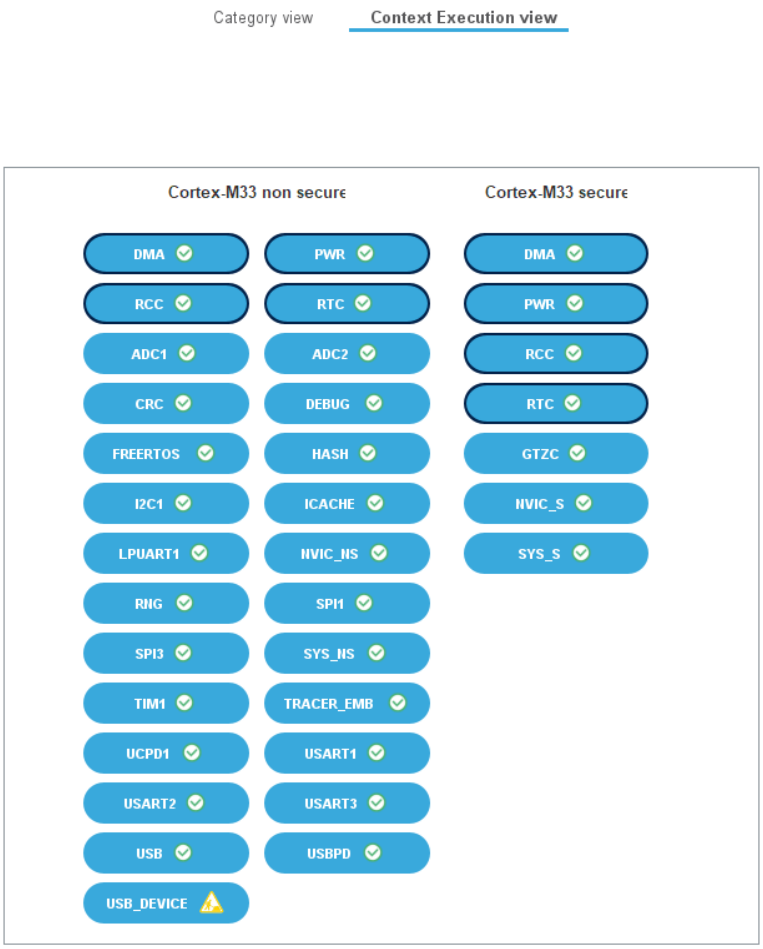
9. System Views

9.1. Category view

9.1.1. Current



9.2. Context Execution view



10. Docs & Resources

Type	Link
Datasheet	http://www.st.com/resource/en/datasheet/DM00532748.pdf
Reference manual	http://www.st.com/resource/en/reference_manual/DM00346336.pdf
Programming manual	http://www.st.com/resource/en/programming_manual/DM00650283.pdf
Errata sheet	http://www.st.com/resource/en/errata_sheet/DM00537981.pdf
Application note	http://www.st.com/resource/en/application_note/CD00160362.pdf
Application note	http://www.st.com/resource/en/application_note/CD00167594.pdf
Application note	http://www.st.com/resource/en/application_note/CD00259245.pdf
Application note	http://www.st.com/resource/en/application_note/CD00264342.pdf
Application note	http://www.st.com/resource/en/application_note/CD00264379.pdf
Application note	http://www.st.com/resource/en/application_note/DM00042534.pdf
Application note	http://www.st.com/resource/en/application_note/DM00072315.pdf
Application note	http://www.st.com/resource/en/application_note/DM00073853.pdf
Application note	http://www.st.com/resource/en/application_note/DM00081379.pdf
Application note	http://www.st.com/resource/en/application_note/DM00085385.pdf
Application note	http://www.st.com/resource/en/application_note/DM00129215.pdf
Application note	http://www.st.com/resource/en/application_note/DM00257177.pdf
Application note	http://www.st.com/resource/en/application_note/DM00272912.pdf
Application note	http://www.st.com/resource/en/application_note/DM00226326.pdf
Application note	http://www.st.com/resource/en/application_note/DM00355687.pdf
Application note	http://www.st.com/resource/en/application_note/DM00311483.pdf
Application note	http://www.st.com/resource/en/application_note/DM00407776.pdf
Application note	http://www.st.com/resource/en/application_note/DM00380469.pdf
Application note	http://www.st.com/resource/en/application_note/DM00395696.pdf
Application note	http://www.st.com/resource/en/application_note/DM00493651.pdf
Application note	http://www.st.com/resource/en/application_note/DM00535045.pdf

Application note	http://www.st.com/resource/en/application_note/DM00536349.pdf
Application note	http://www.st.com/resource/en/application_note/DM00355688.pdf
Application note	http://www.st.com/resource/en/application_note/DM00476869.pdf
Application note	http://www.st.com/resource/en/application_note/DM00532722.pdf
Application note	http://www.st.com/resource/en/application_note/DM00532735.pdf
Application note	http://www.st.com/resource/en/application_note/DM00625700.pdf
Application note	http://www.st.com/resource/en/application_note/DM00660346.pdf
Application note	http://www.st.com/resource/en/application_note/DM00669465.pdf
Application note	http://www.st.com/resource/en/application_note/DM00625692.pdf
Application note	http://www.st.com/resource/en/application_note/DM00666290.pdf
Application note	http://www.st.com/resource/en/application_note/DM00684243.pdf
Application note	http://www.st.com/resource/en/application_note/DM00532737.pdf